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0 seconds

All right, let's do some gallbladder pancreas fun.

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7 seconds

Um, okay. I'm first going to start with the gallbladder. And I'm going to go back up to this picture here.

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16 seconds

So, this is our hpatobiliary system.

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And just a couple of things, some anatomy I want to point out because

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it's going to be important when I start to talk about these different ducts.

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33 seconds

Um, this is our hippatic duct, specifically your right hippatic, the

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42 seconds

patient's right, not our right. Uh, this is your left hypatic duct, your common hypatic duct.

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the cystic duct that connects to the gallbladder. The big daddy is your common bile duct.

1:00

1 minute

And this is your pancreatic duct. Okay?

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And this is where your bile enters the dadum. Okay?

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1 minute, 15 seconds

So our fat is digested in the diodonum of our small intestine.

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1 minute, 24 seconds

Bile is made in the liver, secreted through these ducts, stored in the gallbladder for when we need it. Uh and

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1 minute, 31 seconds

when our dadum senses that there is fat to be digested, our gallbladder is going to release some bile so that we can digest those fats.

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Okay?

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So let's talk about our gallbladder issues.

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So there is [clears throat] excuse me actually I'm gonna I'm going to go here first. There's cholethiasis and there's

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1 minute, 57 seconds

choleiccyitis and cholethasis just like nephro lethasis.

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So nephro kidney lethiosis stone kohi gall. So these are gallstones.

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Okay. And these are what gallstones look like. Isn't that yummy? That is

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incredibly disgusting. Um, except that normally we're not going to have like

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this many gallstones. I mean, like that's pretty impressive. Um, both of these pictures, that's a ton of gallstones.

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But our gallstones are pretty much made kind of with the same idea of how our kidney stones are made.

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So, um, our bile consists of

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Billy Rubin, uh, cholesterol. Oops.

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Sorry, I'm doing this outside, so uh, I don't have a whole lot of room and

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I'm sorry. um Billy Rubin cholesterol, calcium salts,

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and there's a formation of what we call bile

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salts, which is the bio salt is what we actually need to uh in the bile itself to absorb those fats. So, how do we get gallstones?

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except it's primarily the same kind of pathophysiology as our kidney stones. So

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either we have very high concentration of something in the case of gallstones it's a cholesterol thing.

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Okay. So your gallstones can form from very very high concentration of cholesterol or when we have decreased bile salts.

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Okay.

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So, the other, you know, like in kidney stones, I was

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4 minutes, 14 seconds

like, well, there's solutes and then there's this like deficit of water to like kind of flush that out and break up

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those solutes. Kind of again, if we want to think of our solutes like high concentration of cholesterol,

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low bile salts, but then we also have something called biliary

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um sludge or when our bile is very sluggish.

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Okay, so slowm moving thick bile, biliary sludge, think of it like very sluggish. So kind of same concept as

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4 minutes, 52 seconds

blood. When blood sits, blood clots and same with your bile. Okay, so one of those two things is going to cause our

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gallstones. So actually kind of one of three things. So increased cholesterol,

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decreased bile salts or sluggish bile flow.

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So let's talk about our risk factors.

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Okay, this is my absolute favorite nursing trick.

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Um, and here you go. It's something I still remember from nursing school and that was

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oh a hot minute ago 12 years ago. Okay. So this is the four

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Fs. So our risk factors for gallstones is four Fs. Female,

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fat, fertile, and 40.

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So those are our risk factors for developing gallstones.

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Obviously we can kind of connect this one with a high concentration of

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cholesterol. Okay. Um but why female fertile 40?

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Uh, well, fertile because if we've given birth to multiple children, it puts us

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at risk. I'm not entirely sure like what's going on with that, but it does.

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Um, estrogen supplementation too, PO contraceptives.

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So, if we are in the fertile age like 40. So, they just made this uh little saying, the four Fs, female, fat,

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fertile, 40. Okay, you guys will remember it forever.

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[clears throat]

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And oddly enough, like as something that I remember very distinctly being taught in nursing school, a female

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fat fertility, it's like always nothing that comes to mind when seeing a patient in like severe pain that is of

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age and have had multiple children and on the little bit of a

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7 minutes, 14 seconds

obese side of the the BMI chart. I don't know why I just don't ever think of it when I know this little saying so well.

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But anyways, so let's talk about your signs and symptoms. Okay, when we have gallstones, it is very very similar to nephrolithis. So it's like you know th

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these pictures are quite severe. But let's just say this patient has like two gallstones in their gallbladder. Kind of like if they have two little stones in

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their kidney, are they going to cause any symptoms?

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I mean not really. like they could, but like for the most part, if the patient has two little gallstones in their

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7 minutes, 49 seconds

gallbladder, it's more like an in incidental finding um like on a scan and we're like, "Oh,

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7 minutes, 57 seconds

well, also you have some gallstones in there." And they're like, "Oh, really?" But it has like nothing to do with their signs and symptoms and doesn't really explain the reason why the patient is coming in for XYZ, whatever.

8:08

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So, [clears throat] very similar to kidney stones in that regard.

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Um gallstones cause a problem if uh yeah I mean your gallbladder is like that and when we start to develop obstruction

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so that is the main problem with our gallstones. Okay.

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So let's go up here to choleiccyitis. So how do we get from choo lethasis

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to kcyitis is really now this is gallbladder inflammation. So now our number one reason for choleiccyitis is gallstones.

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8 minutes, 46 seconds

Okay?

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Because we're going to have these gallstones as you see here in this gallbladder. They're going to lodge here like a kidney stone. Think like kidney

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8 minutes, 55 seconds

stone a urer. Now we've got gallstones in the bile ducts. Okay? And it's going to cause some pain, some inflammation of

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our gallbladder because kind of the same premise as kidney stones. kidney stones.

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9 minutes, 9 seconds

We've got a stone here. It's going to back up. Urine's going to back up. Same idea here. If we have a stone here, our

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9 minutes, 18 seconds

bile can't escape and it's going to back up and cause inflammation in our gallbladder. So, that is choleiccyitis.

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So, what signs and symptoms are associated with our patients with gall stones and choleiccyitis?

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9 minutes, 32 seconds

Um well just like our kidney stones that have renal collic our patients have something called biliary collic

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and biliary collic is severe spasms of the right upper quadrant. So your gallbladder sits like well if we go back

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9 minutes, 49 seconds

up to that other picture your liver is kind of like right here. Okay so your liver is right upper quadrant. So your gallbladder is kind of in that area too.

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Right upper quadrant. Okay. So biliary collic is severe spasms of the right upper quadrant

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and kind of like we talked about in bowel obstructions in mechanical bowel obstructions we have like this increased paristalsis to try

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to like move that obstruction. Well that's exactly what's happening in our gallbladder. Our gallbladder is like contracting and trying to move bile past

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the stone. It's just unable to because it's lodged in this duct. So that is how we get bilary collic.

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The pain in um choleiccyitis, this bilary collic is usually sharp

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and it radiates oddly enough to the right shoulder.

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Okay. Their pain is worse after a fatty meal.

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Oh my gosh, that should make so much sense if we know numero uno. When bile gets released is when our datadum senses

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that we have fat that we need to break down. So our gallbladder contracts, releases this bile. So after we have a fatty meal, our dad's like, "Hey, I need

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to digest these fat molecules." And our gallbladder is like, "I'm trying to help you, but the stone is like he

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uh yeah, we can't get there. I can't let the bile through because of this stone." So that makes sense why our pain is worse after a fatty meal.

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Um, nausea, vomiting, yeah, pain response, abdominal issue. That should make sense.

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Let's talk a little bit about these other problems here. So the patient gets jaundest

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and they might have a little bit of

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dark urine. Okay, so their Billy Rubin is going to be up. Okay, so let's think about this.

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If my stone is right, Actually, let me go back up to this other picture.

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because it says all of them. Okay. If let's say my stone is right here.

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[clears throat]

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Okay. My stone's right here. Not letting any bile get by. Okay.

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Well, my liver, all this bile is going to back up here

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and here, inflame my gallbladder. Number one, choleaccyitis.

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But number two, it's also going to cause issues in my liver. So my liver usually takes that unconjugated Billy Rubin,

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transforms it to conjugated Billy Rubin, puts in the bile. Well, when these ducks are overflowing with bile, my liver's like, "Okay, well clearly I don't think

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12 minutes, 51 seconds

I need more Billy Rubin, so I'm just going to let you hang out in the blood because I don't really know what else to do with you." Okay, so we got

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13 minutes

unconjugated Billy Rubin or Billy Rubin that we can't really use sitting out in the blood. It's going to raise our belly ribbon levels and these patients are going to be yellow or jaundice. Okay. Uh

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the other thing that might happen is dark urine.

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Okay. And this happens because we're trying to get rid of the Billy Rubin somehow. So we're going to see Eurobilogen.

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I believe that's how you say it. And I believe that's the proper term. Eurobilogen.

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Okay, whatever. Fancy term for Billy Rubin in in the pee. Okay, because my body's like, I need to get

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13 minutes, 41 seconds

rid of it somehow. I guess I'll just pee it out. Maybe since my liver is not really working, doing its job. H interesting. Okay.

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Um, with choleiccyitis, let's go back down here. Lucyitis that can also because this is an inflammatory

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process like an appendicitis, fever, increased white blood cell count or luccoytosis.

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Um you're not going to see that with like your unobstructed gallstones.

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Okay. So this disorder leads itself to what is known as pancreatitis also. So same picture here.

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Um, if we have a stone that is blocking our common bile duct way down here,

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okay, then it's not only going to affect my gallbladder, my liver, but also now my pancreas and my pancreatic duct.

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Okay.

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So, what happens? Bile is backs up to my liver. Bile backs up in my gallbladder

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causes choleaccyitis. And now I can't really do anything with these pancreatic

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enzymes. So our pancreas really doesn't have anything to do with bile. Like it is ducks and they are pictured as green

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down here. I mean that it kind of does hit this area a little bit but um like I

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15 minutes, 15 seconds

don't want you guys to think like oh the pancreas is involved in bile production or storage of bile. So that's our liver and our gallbladder. Our pancreas is

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just right next door. And our pancreas has endocrine function, which we'll get into next week, and exocrine function.

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So the exocrine function of the pancreas is pancreatic enzymes.

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And our pancreatic enzymes are very important in the digestive process. Um

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we've got trison, amalase, lipase. Okay.

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15 minutes, 50 seconds

Um, I know lipase helps digest fats.

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15 minutes, 57 seconds

I don't know exactly what these two do, but lip lipo

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fats. Anyways, these are we have lab values for ama and lipase. whenever

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we're kind of looking at the uh gallbladder, pancreas, we are looking at lipase levels also. So

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anyways, lipase is a pancreatic enzyme that gets just released through these ducts to go to the dadum to absorb nutrients. So if I have a gall stone

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16 minutes, 30 seconds

sitting here, then my pancreatic enzymes can't escape.

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16 minutes, 35 seconds

And if my pancreatic enzymes can't escape, well, um, they're going to start to build up

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in here and kind of just like we call it auto digestion

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16 minutes, 51 seconds

of the pancreas. So, they kind of start to eat away at the pancreas itself. Okay.

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16 minutes, 59 seconds

Um, yeah, that's not really good.

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So when I think pancreatitis, I just kind of think sepsis and this is how we treat it in case anyone was wondering.

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Although we don't talk about treatment in this class, but um these patients need like fluids and to fix the problem of causing their pancreatitis. Okay, cool.

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17 minutes, 25 seconds

Um, also maybe going to have an increased white blood cell count, maybe a fever, but the main sign and

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symptom these patients are going to experience is pain because the pancreas is literally eating itself and that is a very painful process and then we're

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17 minutes, 41 seconds

going to have increased lip pace. Um, what else? Oh, I was going to say if it's not a gall stone that's causing

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17 minutes, 49 seconds

pancreatitis, that's kind of how we like jumped to this topic because absolutely it can cause pancreatitis. But our other

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17 minutes, 56 seconds

main cause of pancreatitis is alcohol, especially binge alcohol is

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going to cause um acute pancreatitis and

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the way that alcohol causes pancreatitis. So gallstones is obviously an obstruction issue, but alcohol,

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18 minutes, 18 seconds

there's toxic byproducts that kind of just make the pancreas very unhappy and starts to like say, "Okay, yep, all

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18 minutes, 27 seconds

these enzymes can now work in my pancreas. I don't need them to go to the dadum to work." So that's kind of how

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18 minutes, 34 seconds

pancreatitis works in the setting of um an acute alcohol binge.

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18 minutes, 41 seconds

Um, let's talk about some complications.

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18 minutes, 46 seconds

Oh, you know what? Actually, I'm not really done with our signs and symptoms. Okay, signs and symptoms pancreatitis.

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18 minutes, 52 seconds

So, their pain is going to be epigastric and it's going to be increased with the supine position. So, when they're laying

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19 minutes

flat on their back, they are going to have more pain.

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Their pain can radiate to the back also.

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Let's talk about complications. I think that's all I really wanted to say about pancreatitis.

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Okay, complications.

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So, the pancreas is kind of an interesting abdominal organ. Most of your abdominal organs like your liver, your kidneys, if you want to say that's

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abdominal, it's retroparitinal, but whatever. Um, they have capsules around them and the capsules are kind of

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protective to like localize any problem that happens in those organs and it's pretty cool little mechanism they got going there. Uh, our pancreas not so

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lucky. So when our pancreas gets inflamed, we have no protective capsule to keep it localized. So these enzymes

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20 minutes

this mass inflammation likes to spread to adjacent tissues and one of the things that we can see so obviously

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these patients are very high risk for sepsis infection. Okay. Um because you know all this inflamed tissue is very

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easy for just any sort of bacteria to come and make its home in a moist adematous environment.

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Um, but one of the other places that the pancreatic enzymes like to kind of go and this inflammation likes to spread is

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actually our lungs because our lungs are not too far from this whole area. And it's very interesting that kind of like crosses the diaphragm and whatnot goes

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into the lungs. So, your patients, even though it's kind of weird to think about, um, your patients with acute

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pancreatitis can absolutely have like secondary lung infections or lung inflammation.

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Um, yeah, very interesting. Plural eusions.

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21 minutes, 4 seconds

It's very odd. Very interesting. Um, the other thing that it can cause because it doesn't have a capsule around it is we

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don't really need this pancreas to burst. That would be horrible. And that doesn't really happen. Um,

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but we don't have this capsule around it. So, if this thing is going to be massively inflamed and angry because

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21 minutes, 28 seconds

it's being autodigested by its own enzymes, then we at very high risk for peritonitis because this thing sits in the paritinium just kind of all by

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21 minutes, 35 seconds

itself. nothing around it. So any little inflammation, any little bacteria that kind of sits in here can easily spread to the paritinium and we can develop peritonitis.

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21 minutes, 49 seconds

Obviously necrosis and eskeemia is always a risk with any like widespread inflammation because it's pressing on those vessels and causing them to not be

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perfused as well. So, um, again, not that we really talk treatment, but I

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just do want to mention like, you know, uh, the way that we kind of treat pancreatitis is like we just don't let

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these patients eat, which is very sad, but like we don't want anything getting in this datadum to like stimulate this pancreas of like,

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22 minutes, 21 seconds

oh, I need to produce enzymes and the enzymes can't get out and then uh, yeah, the pancreas gets worse. So, we just kind of keep them NPO.

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Um, oh, let's also talk about this real quick. If this is the main kind of

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highway for things, I also do just want to touch on this endocrine component of the

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22 minutes, 46 seconds

pancreas. So, um, if our pancreas is like super inflamed and not really working that well,

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um, yeah, our blood sugar regulation isn't going to be the best because

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that's our endocrine function of um, our pancreas.

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So, just keep that in mind.

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And that is pretty much it about pancreatitis and gallstones and choleiccyitis. I have fun questions in

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23 minutes, 20 seconds

here. If you guys want to do them, let me know what you think.